RAJALAKSHMI ENGINEERING COLLEGE

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(CYBER SECURITY)

CR23532 – CYBER AND DIGITAL FORENSICS
CASE STUDY ASSIGNMENTS – II

CONSOLIDATED SUBMISSION
(All Five Case Studies – Questions 1 to 38)

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Submitted in partial fulfillment of the coursework assessment requirements for CR23532 – Cyber and Digital Forensics (Academic Year 2026–2027, Odd Semester).

CONTENTS OF THE CONSOLIDATED SUBMISSION

Each case study follows the same structure — case overview with facts separated from assumptions, the numbered assignment questions with their deliverable tables and figures, limitations, conclusion and references. All case data is hypothetical and is labelled as such; no forensic tool was executed for this coursework.

Case Study	Title	Questions	Scenario ID
2(1)	Suspected Unauthorized Access to a College Server	Questions 1–7	REC-DFIR-2026-01
2(2)	Linking a Suspicious File to a User Using File-System Evidence	Questions 8–14	REC-DFIR-2026-02
2(3)	Forensic Imaging and Safekeeping of a Suspect Storage Device	Questions 15–22	REC-DFIR-2026-03
2(4)	Recovery of Deleted Evidence from Volume Shadow Copies	Questions 23–30	REC-DFIR-2026-04
2(5)	Multi-Source Investigation of a Suspected Cyber Incident	Questions 31–38	REC-DFIR-2026-05

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Note on formatting: one front page serves the whole consolidated submission — the college logo and the candidate particulars appear once, here at the top. Each case study then starts on a new page, tagged in the top margin of its opening page, and keeps its own footer page numbering (2, 3, …) so the five reports remain independently referenceable against the separate submissions.

1. CASE OVERVIEW, KNOWN FACTS AND ASSUMPTIONS

A laboratory administrator reports that confidential end-semester question paper drafts stored on the departmental examination server (CSE-EXAM-SRV01) may have been read and copied outside working hours. The college requires an investigation that is technically sound, legally defensible and administratively usable. This report answers Questions 1–7 of Assignment 2(1) and produces the four required deliverables: an investigation flowchart, an evidence classification table, the case analysis, and a sample forensic report structure. No forensic tool was executed for this coursework; all workflows are described conceptually and all case data below is hypothetical.

Table 1.1: Separation of Known Facts, Hypothetical Details and Analytical Assumptions

Category	Details
Known facts (from the assignment brief)	Suspected unauthorized access to a college server; an initial complaint exists; a laboratory workstation and a USB device are within scope; server access logs and confidential examination files are involved.
Hypothetical case details	Incident window 12 Oct 2026, 22:10–22:35 IST; suspect workstation CSE-LAB3-PC09; shared maintenance account lab_maint; hidden share \\COE_Papers_Odd2026$; files CS501_QP_Draft.pdf and CS503_Key.pdf; 32 GB SanDisk Ultra USB, serial 4C530001205118109342.
Analytical assumptions	Written authorization from the Principal / Controller of Examinations is obtained; log rotation on the server is frozen before export; the workstation was found powered on; NTP synchronization status of each device is measurable and recorded.
Excluded from scope	Identification of the human being at the keyboard, motive, and any disciplinary verdict — these rest with the institutional authority, informed by biometric or CCTV corroboration.

2. DETAILED ANSWERS TO THE ASSIGNMENT QUESTIONS (QUESTIONS 1–7)

2.1 Five-Phase Forensic Investigation Lifecycle (Question 1)

Question 1: Explain how the forensic team should move through the Identification, Collection, Examination, Analysis, and Presentation phases for this case.

The team follows the five-phase lifecycle recommended by NIST SP 800-86 and ISO/IEC 27037, with each phase gated by a documented output that the next phase consumes. No phase may be entered until the previous phase's output has been verified and recorded.

Table 1.2: Five-Phase Lifecycle Applied to the College Server Case

Phase	Key Actions in This Case	Gate / Output
Identification	Receive the administrator's written complaint and define the incident window; confirm signed institutional authorization; catalogue evidence repositories — the complaint and incident notes, the lab workstation (RAM and storage), server authentication and SMB audit logs, the physical USB device, the target examination files — plus corroborating sources such as Lab 3 biometric door logs and DHCP/switch-port mappings.	Signed authorization memo; scoped incident intake form; evidence source inventory

Phase	Key Actions in This Case	Gate / Output
Collection	Photograph screen state, cabling and the inserted USB; record BIOS/OS clock offset against an NTP reference; capture a raw memory image first because the machine is live (Order of Volatility, RFC 3227); then power down and acquire bit-stream images through hardware write-blockers; export server logs read-only. Assign an Evidence ID, hash with SHA-256, seal and log every item.	Sealed exhibits; RAM and disk images with hash values; chain-of-custody entries
Examination	Work only on verified read-only working copies; apply NSRL known-file filtering; parse the NTFS $MFT and $UsnJrnl, Security/System .evtx logs, SYSTEM/SAM/NTUSER.DAT/UsrClass hives, .LNK files, JumpLists and ShellBags; inspect the USB FAT directory entries and carve unallocated clusters for deleted examination files.	Verified artifact extraction set (never the originals)
Analysis	Normalize all timestamps to UTC and IST after applying measured drift; correlate logon (4624 Type 2), share mount (5140/4663), USB arrival (USBSTOR serial match) and file creation; compare SHA-256 of the server files against files or carved remnants on the USB; test counterfactuals such as remote RDP pivot or pre-existing files.	Unified master timeline; hypothesis evaluation matrix
Presentation	Deliver a layered forensic report — plain-language executive summary followed by technical detail — with hash manifests, custody records and an explicit statement of limitations, briefed to the Principal, Controller of Examinations and Disciplinary Committee.	Final forensic report; briefing record

2.2 Forensic Investigation Flowchart (Question 2)

Question 2: Prepare a forensic investigation flowchart showing the sequence of activities from the initial complaint to presentation of findings.

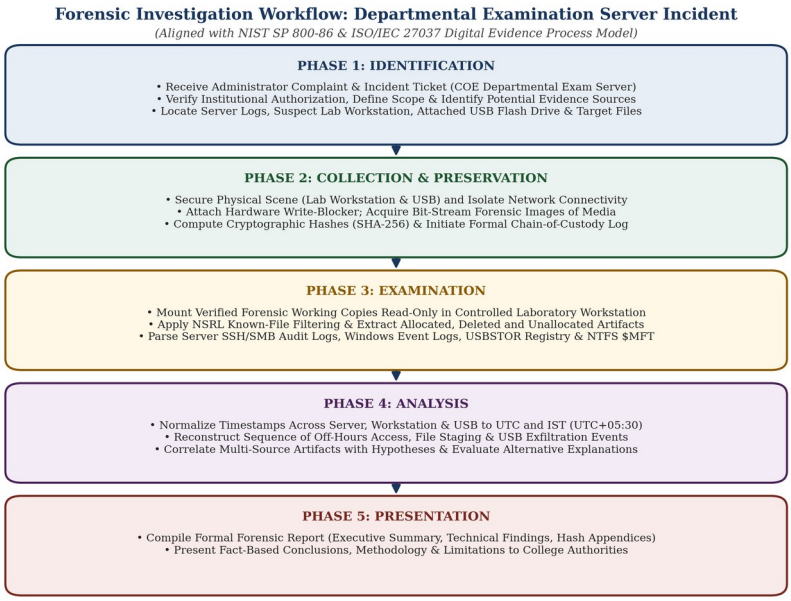


Figure 1.1: Investigation Flowchart — Initial Complaint to Presentation of Findings

Table 1.3: Operational Stage Breakdown Supporting the Flowchart

Workflow Stage	Core Forensic Actions	Verifiable Output
1. Complaint & Authorization	Log incident time; freeze server log rotation; cordon off the Lab 3 terminal; obtain written mandate.	Authorization memo; incident intake form
2. Scene Triage & Volatile Capture	Photograph physical state; measure clock drift; capture RAM image on the powered-on workstation.	Photo log; clock offset record; RAM image + SHA-256
3. Non-Volatile Acquisition	Attach hardware write-blockers; create bit-stream images of SSD and USB; export server logs read-only.	Master images (.E01/.ASB); hash log; custody form
4. Quality Gate (Integrity)	Compare source and image digests; seal originals in the evidence vault. No parsing proceeds on failure.	Verified working copies only
5. Artifact Extraction	Parse SMB/authentication logs, .evtx, registry USBSTOR, $MFT, $UsnJrnl, .LNK/JumpLists; carve deleted files.	Artifact tables; carved files; hash comparisons
6. Cross-Source Analysis	Normalize to UTC/IST; correlate IP, SID, USB serial and file hashes; test alternative hypotheses.	Unified timeline; hypothesis matrix
7. Reporting & Briefing	Draft executive summary and technical annexures; brief non-technical authority; archive the package.	Defensible final report; archival package

2.3 Identification and Classification of Digital Evidence (Question 3)

Question 3: Identify at least five items of potential digital evidence in the case and classify the evidence according to its investigative value.

Table 1.4: Evidence Classification by Investigative Value (Eight Candidate Items)

Item of Digital Evidence	Form / Source	Evidence Class	Investigative Value & Interpretive Caution
Server authentication log	/var/log/auth.log or Security .evtx on CSE-EXAM-SRV01	Direct – system-generated	High: places an account and source IP on the examination share at a specific time. Proves account use, not human identity.
SMB file-access audit record	Event ID 5140 / 4663 or Samba vfs_full_audit	Direct – system-generated	High: names the exact confidential file read or written across the share.
Workstation volatile memory	Raw RAM capture of CSE-LAB3-PC09	Transient – must be captured first	Medium-high: reveals live SMB session handles, mounted paths, open files and residual keys; easily destroyed by shutdown.
Workstation NTFS volume ($MFT, $UsnJrnl)	Bit-stream image of the internal SSD	Direct / linking	High: MACB timestamps evidence local file creation, renaming and deletion.
Windows Registry USBSTOR key	SYSTEM hive	Supporting – device linkage	High for device association: ties the seized USB serial to the workstation and gives first/last connect times.
USB flash drive file system	32 GB SanDisk Ultra, FAT32/exFAT	Direct – storage of copy	High: presence, creation and (if deleted) carving remnants of the examination files; hashes must match server copies.
Lab 3 biometric door log	Access-control controller export	Supporting – physical corroboration	Medium: corroborates who was physically present; separate custodian, so must be requested formally.
Administrator's complaint and incident notes	Written report, helpdesk ticket	Contextual – leads only	Low as proof, high as scoping: initiates the case and defines the window, but is not independent evidence.

2.4 Collection and Preservation Precautions (Question 4)

Question 4: Explain the precautions required during collection and preservation so that the evidence remains reliable and defensible.

Digital evidence is fragile: booting a suspect machine, opening a folder on the USB drive, or copying files with ordinary utilities rewrites access timestamps, creates registry entries and can overwrite deleted artifacts in unallocated space. Seven controls address that fragility.

- **No live browsing of original media.** Examination occurs only on verified working copies; touching the exhibit in Windows Explorer contaminates MACB timestamps and writes System Volume Information metadata.
- **Hardware write-protection.** A validated write-blocker (e.g. Tableau T8u for USB, T35u for SATA/NVMe) sits between suspect media and imaging station and rejects write commands at the bus level.
- **Order of Volatility (RFC 3227).** Because the workstation was powered on, RAM and active network connection tables are captured before isolation and shutdown.
- **Bit-stream rather than logical copy.** A sector-by-sector image (E01 or ASB) preserves deleted records, file slack, unallocated clusters and $MFT entries that file copying discards.

- **Hash verification before and after.** SHA-256 (with MD5 for legacy compatibility) is computed on the source during acquisition and verified against the image; matching digests are mathematical proof that no bit changed.
- **Log preservation out-of-band.** Server authentication and SMB audit logs are exported to write-once media before rotation overwrites them, hashed, and recorded together with NTP synchronization status.
- **Unbroken chain of custody.** Every item is bagged in tamper-evident anti-static packaging, sealed with dated and signed tape, and each transfer is logged contemporaneously in the custody register.

2.5 Establishing a Timeline of the Suspected Access (Question 5)

Question 5: Describe how investigators can establish a timeline of the suspected unauthorized access using available evidence.

- **Step 1 — Normalize the clocks.** Measure the BIOS/OS offset of each device against a Stratum-1 NTP reference. NTFS $MFT and .evtx store UTC FILETIME (100 ns intervals from 1 Jan 1601); FAT32/exFAT on USB typically stores local time with no offset; Linux syslog may hold local IST. Convert everything to UTC with IST (UTC+05:30) shown alongside, and state the residual uncertainty per source.
- **Step 2 — Extract anchor events.** Workstation: 4624/4634 logon-logoff, USBSTOR arrival, DriverFrameworks 2003/2100, .LNK and JumpList creation. Server: 4624 Type 3 network logon, 5140 share access, 4663 object access, or equivalent Samba audit lines. USB: directory-entry creation and modification times.
- **Step 3 — Align causally.** Test that the ordering is physically possible: local logon → server authentication → share mount → file read → device insertion → file creation on USB → removal → logoff.
- **Step 4 — Synthesize and flag gaps.** Publish the timeline in figure and table form, with each event traced to a named artifact and a confidence note where clocks disagree.

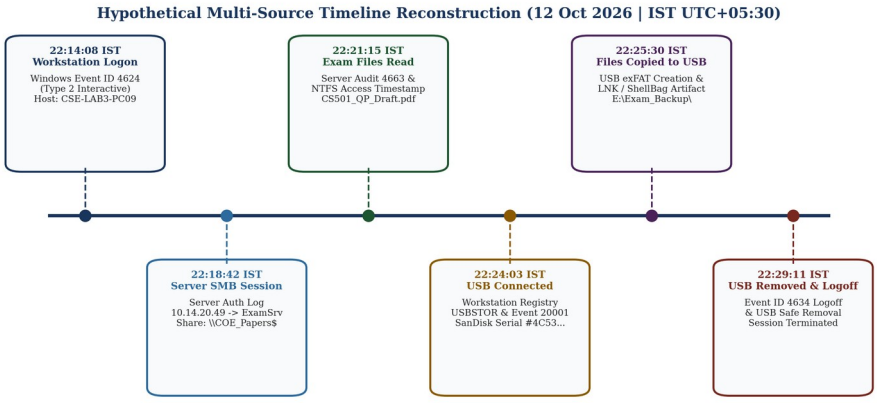


Figure 1.2: Multi-Source Timeline Reconstruction for the Hypothetical Incident Window (12 October 2026)

Table 1.5: Hypothetical Multi-Source Master Timeline

Timestamp (UTC / IST)	Source	Artifact Observed	Interpretation
12 Oct, 16:40:12 / 22:10:12	Lab 3 door controller	Badge swipe entry #88412	Physical entry outside working hours (contextual anchor)
12 Oct, 16:44:08 / 22:14:08	CSE-LAB3-PC09 Security.evtx	Event 4624, Type 2 interactive, lab_maint	Local keyboard session begins on the suspect workstation

Timestamp (UTC / IST)	Source	Artifact Observed	Interpretation
12 Oct, 16:48:42 / 22:18:42	CSE-EXAM-SRV01 auth/SMB log	Session setup from 10.14.20.49 to \ \COE_Papers_Odd2026$	Links that workstation to the examination share
12 Oct, 16:51:15 / 22:21:15	CSE-EXAM-SRV01 audit log	Object read of CS501_QP_Draft.pdf, CS503_Key.pdf	Confidential files were read across the share
12 Oct, 16:54:03 / 22:24:03	Workstation registry	USBSTOR arrival, SanDisk 4C530001205118109342 → E:	The exact seized USB device was connected
12 Oct, 16:55:30 / 22:25:30	USB directory + .LNK	E: \Exam_Backup\CS501_QP_ Draft.pdf (SHA-256 = server copy)	Examination file copied to USB and opened locally
12 Oct, 16:59:11 / 22:29:11	Workstation Security.evtx	Event 4647/4634; USB left in port	Matches the locked state found the next morning

2.6 Final Report Content and Briefing a Non-Technical Authority (Question 6)

Question 6: Explain what should be included in the final forensic report and how findings should be presented to a non-technical authority.

The report serves a dual audience: institutional decision-makers who must understand what happened and how reliable the proof is, and technical peers or counsel who may audit the methodology. Table 1.6 is the deliverable sample report structure.

Table 1.6: Deliverable Sample Forensic Report Structure

Report Section	Required Content	Value to the Reader
1. Cover & metadata	Case ID, title, team, authorizing authority, date, classification.	Document control and accountability
2. Executive summary	One to two plain-language pages: background, scope, short timeline, key conclusions, policy impact.	Non-technical readers grasp findings without log dumps
3. Scope & authority	The written authorization and the specific questions answered.	Shows collection complied with governance and privacy rules
4. Evidence received & custody	Inventory of every exhibit with Evidence ID, serial, hash and transfer log.	Demonstrates continuity of possession
5. Methodology	Write-blockers, imaging tooling, pre/post SHA-256 verification, laboratory workflow.	Repeatability and admissibility (BSA 2023 s.63)
6. Technical findings	Correlation of SMB logs, event and registry artifacts, USB metadata, hash matches.	Verifiable proof behind every stated fact
7. Chronological timeline	Normalized UTC/IST table plus the timeline figure.	Minute-by-minute sequence for the committee
8. Limitations & alternatives	Shared credentials, possible remote-access vectors, surveillance gaps.	Prevents over-attribution; shows impartiality
9. Conclusions & remediation	What is proven versus unproven; hardening recommendations (SMB ACLs, USB port control).	Enables action and recurrence prevention
10. Appendices & glossary	Full hash manifest, custody forms, s.63-style certificate, plain-English glossary.	Complete supporting documentation

Presenting to a Non-Technical Authority

- **Layer the communication.** Open with the executive summary and the timeline figure, keep registry paths and hex offsets in referenced annexures, and let the committee read detail only if it challenges a conclusion.
- **Use accurate analogies.** Describe a SHA-256 digest as a tamper-proof digital fingerprint proving two files are identical, and a write-blocker as a one-way barrier that permits reading while physically preventing change.
- **Separate fact from inference.** State what the data proves — account lab_maint on PC09 copied CS501_QP_Draft.pdf to the SanDisk drive at 22:25 IST — and what it does not — which person typed at the keyboard, which needs biometric or CCTV corroboration.
- **Stay impartial in tone.** The investigator is a finder of technical fact, not an advocate; avoid language about motive or guilt.

2.7 Legal and Ethical Considerations (Question 7)

Question 7: Briefly discuss the legal and ethical considerations involved in collecting digital evidence in a college environment.

- **Ownership versus privacy.** The workstation and server are institutional property governed by the Acceptable Use Policy, but they may hold unrelated personal data of students and staff. Investigators should confirm the AUP gives notice of monitoring and audit, and obtain written authorization from the head of the institution before seizure.
- **Minimization and proportionality.** Search strictly within the relevant date range, filenames and device artifacts; indiscriminate exploration of unrelated personal files is unethical and may be unlawful.
- **Statutory compliance.** Unauthorized access to a computer resource and copying data without permission engage Sections 43 and 66 of the Information Technology Act, 2000, with the Digital Personal Data Protection Act, 2023 governing personal data handling. Where the matter may reach a court, electronic records must satisfy Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 (successor to s.65B of the Evidence Act), which requires a certificate on proper device operation — reinforcing hashing and documented custody.
- **Confidentiality and natural justice.** Leaking question paper contents or a suspect's identity before verification causes irreparable academic harm; findings go only to the authorized committee.
- **Conflict of interest.** The administrator who reported the incident and custodies the server should not be the sole examiner of those logs; independent review preserves objectivity.

3. LIMITATIONS AND ALTERNATIVE EXPLANATIONS

Table 1.7: Attribution Limits That Constrain the Conclusions

Limitation	Effect on Interpretation	Corroboration Required
Shared laboratory credentials	Artifacts prove which account and machine acted, not which person sat at the keyboard.	Lab biometric door logs, CCTV, roster and supervisor statements
Possible remote pivot	If RDP, VNC, SSH or a remote access tool was active on PC09, an external actor could have reached the server through it.	Logon Type in Event 4624 (2 interactive vs. 10 remote vs. 3 network) plus firewall and NetFlow records
FAT32 timestamp weakness	USB access times carry no time zone and only one-day resolution, so they cannot alone fix a time.	Corroborate against NTFS .LNK and USBSTOR UTC records from the workstation

Limitation	Effect on Interpretation	Corroboration Required
Server log integrity	Logs may be rotated, filtered or administratively edited before export.	NTP status, log rotation configuration, hash of raw export, retention policy review

4. CONCLUSION

Applied to this case, the five-phase lifecycle lets investigators reconstruct an off-hours data breach while leaving every original byte untouched: write-blocked bit-stream imaging, SHA-256 verification and UTC-normalized correlation of server SMB logs, workstation event and registry artifacts, and the USB file system produce a chain of events that is mathematically checkable rather than merely plausible. On the hypothetical data presented, the evidence supports the conclusion that the confidential examination files were read from the hidden share and copied to the specific seized USB device within the reported window. It does not, by itself, establish which individual performed the act, and that limit is stated openly in the report. Presented through a layered, jargon-free structure with full custody documentation, these findings allow the institution to take defensible disciplinary and remedial action — notably SMB access-control hardening, disabling idle network shares, and enforcing USB port control.

5. REFERENCES

[1] Kent, K., Chevalier, S., Grance, T., & Dang, H. (2006). Guide to Integrating Forensic Techniques into Incident Response (NIST SP 800-86). National Institute of Standards and Technology.

[2] International Organization for Standardization. (2012). ISO/IEC 27037:2012 — Guidelines for identification, collection, acquisition and preservation of digital evidence.

[3] Brezinski, D., & Killalea, T. (2002). Guidelines for Evidence Collection and Archiving (RFC 3227, BCP 55). Internet Engineering Task Force.

[4] Carrier, B. (2005). File System Forensic Analysis. Addison-Wesley Professional.

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[6] Government of India. (2023). The Bharatiya Sakshya Adhiniyam, 2023 (Act No. 47 of 2023), Section 63: Admissibility of Electronic Records.

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1. CASE OVERVIEW, FACTS AND ASSUMPTIONS

A desktop computer shared by several staff members contains multiple user accounts along with a suspicious executable, hidden files, encrypted documents and browser artifacts. Investigators suspect that one particular account created and accessed those files, and must document the environment, design a file-system analysis workflow and test whether the artifacts really tie the activity to that account. This report answers Questions 8–14 and delivers the four required outputs: a system inventory table, an evidence-linkage matrix, an ILookIX analysis workflow, and a conclusion with its limitations. ILookIX is described conceptually from syllabus principles — no tool was executed, and every path, SID and timestamp below is hypothetical.

Table 2.1: Known Facts, Hypothetical Scenario Details and Working Assumptions

Category	Details
Known facts (assignment brief)	Shared multi-user Windows desktop; a suspicious executable, hidden files, encrypted documents and browser artifacts are present; the objective is to link file creation/access to one user account.
Hypothetical case details	Hostname FIN-DESK-04, four staff accounts s_meena (RID 1002), r_kiran (1004), v_balaji (1005), admin_support (500/1001); net_diag_helper.exe in C:\Users\Public\Documents\Diag\; hidden .stashed_records.db; ADS notes.txt:secret.cfg; AES-256 archive Staff_Payroll_Export_2026.7z in r_kiran\Documents\Archive\; activity window 11 Oct 2026, 14:05–14:31 IST.
Working assumptions	Windows 11 Enterprise 64-bit on GPT/NTFS 3.1 NVMe SSD with a secondary SATA HDD; no attribution is drawn from a single artifact; multi-artifact corroboration across independent OS subsystems is mandatory before any conclusion is stated.

2. DETAILED ANSWERS TO THE ASSIGNMENT QUESTIONS (QUESTIONS 8–14)

2.1 Hardware and Software Components to Document Before Analysis (Question 8)

Question 8: Identify the hardware and software components that should be documented before analysis.

Documenting hardware ensures no storage medium, hidden partition or attached peripheral is overlooked; documenting software establishes how the operating system generates, alters and purges file-system and user-activity artifacts. Table 2.2 is the deliverable system inventory.

Table 2.2: Deliverable Pre-Analysis Hardware and Software System Inventory

ID / Category	Documented Specification (Hypothetical)	Forensic Significance
HW-01 Chassis, board, firmware	Dell OptiPlex 7090, asset REC-FIN-04; UEFI v1.24.0; Secure Boot state; CMOS clock compared to a Stratum-1 UTC reference.	Machine identity, boot order, and measured clock skew (+42 s fast) needed to normalize every timestamp.
HW-02 Primary storage	WD SN730 512 GB NVMe SSD, S/N 2144E8804912; GPT with EFI, MSR, Windows (C:), Recovery partitions.	Identifies the physical source of the OS, user profiles, hives, browser databases and the NTFS $MFT/$UsnJrnl.
HW-03 Secondary and removable storage	Seagate BarraCuda 1 TB SATA HDD (D:); optical drive; USB controllers and any attached dongles or cards.	Ensures staging volumes, hidden partitions, and Host Protected Area / Device Configuration Overlay regions are imaged and examined.

ID / Category	Documented Specification (Hypothetical)	Forensic Significance
HW-04 NIC and memory	Intel I219-LM Gigabit (MAC 00-1B-63-84-45-E6); 16 GB DDR4 RAM.	Ties the desktop to DHCP and switch logs and sizes the volatile memory capture.
SW-01 Operating system	Windows 11 Enterprise 64-bit, Build 22631.4169; install date; TimeZone key India Standard Time (UTC+05:30).	Determines artifact behaviour, e.g. whether NtfsDisableLastAccessUpdate is set and how Prefetch and Amcache record execution.
SW-02 File system and volumes	NTFS 3.1 on C: and D:, 4096-byte clusters; volume serials 7C94-A12B and 4E81-09F2; BitLocker and VSS status.	Enables .LNK volume-serial correlation, sets cluster-slack expectations, and records whether shadow copies exist.
SW-03 Accounts and privileges	SAM and ProfileList entries: s_meena (1002), r_kiran (1004), v_balaji (1005), admin_support (500/1001), with privilege level and password-reset dates.	Maps each SID to a named staff member and to standard or administrator rights — the basis of any attribution claim.
SW-04 Applications	Chrome 129, Edge, 7-Zip 24.08, Defender logs, RDP/AnyDesk state, any cleanup or wiping tools.	Identifies the software able to create the encrypted archive, download the executable, or access the machine remotely.

2.2 How Storage Devices, Operating Systems, File Systems and Metadata Locate Evidence (Question 9)

Question 9: Explain how storage devices, operating systems, file systems, and metadata can help investigators locate relevant evidence.

- **Storage devices — where traces can hide.** GPT or MBR maps allocated partitions and unpartitioned gaps. Sector-level access exposes allocated clusters, unallocated space holding deleted files until overwritten, file slack between logical end-of-file and cluster end, and SSD wear-levelling or TRIM behaviour that affects recoverability.
- **Operating system — who did what and when.** Windows isolates each profile under C:\Users\<name> and records accountability in the registry: SAM and ProfileList for SIDs and last logon; per-user NTUSER.DAT and UsrClass.dat for RecentDocs, RunMRU, UserAssist (ROT-13 encoded execution counts) and ShellBags (folder traversal even after deletion). Prefetch, Amcache.hve and Security.evtx record binary execution and logon sessions.
- **File system — the ledger of operations.** $MFT holds a record per file and stores small files (under ~700 bytes) as resident data; $UsnJrnl:$J and $LogFile journal creation, modification, renaming and deletion; $I30 index slack retains names of files once in a folder; $Recycle.Bin pairs deleted data ($Ixxxxxx) with metadata recording the original path and deletion time inside the deleting user's SID subfolder.
- **Metadata — the linking layer.** File-system metadata supplies $STANDARD_INFORMATION and $FILE_NAME MACB times, the Owner SID in $SECURITY_DESCRIPTOR, attribute flags, and the Zone.Identifier stream naming the download URL. Application metadata inside Office files, PDFs and executables carries author and last-saved-by names, revision counts, and PE-header compile timestamps.

Together these layers let an investigator move from a named file to a named account: the artifact's path places it in a profile, its Owner SID names the security context, its journal entries fix the time, and execution records in that user's own hives show the account acted on it.

2.3 Hidden and Encrypted Files and Their Evidence-Handling (Question 10)

Question 10: Describe how hidden files and encrypted files may affect a forensic investigation. Discuss appropriate evidence-handling considerations.

Effect on the investigation

Concealment techniques on a shared desktop include setting the NTFS Hidden (0x02) or System (0x04) attributes so items are invisible in Explorer, dot-prefixed names or placement in obscure system folders, extension mismatch (a .7z or .exe renamed to .dll or .tmp), and payload data inside an Alternate Data Stream such as notes.txt:secret.cfg, where the parent file looks benign while a secondary $DATA stream holds the content. These defeat casual administrative review but not forensic parsing of the raw $MFT, which ignores visibility flags and immediately surfaces signature-versus-extension mismatches.

Encrypted content — password-protected AES-256 archives, Office Agile Encryption files, BitLocker or VeraCrypt containers, EFS files — blocks indexing, keyword searching and content reading, and high-entropy data also frustrates signature carving when fragments are scattered. Encryption does not, however, erase the surrounding file-system record: unless the volume itself is unmounted, the name, path, size, $MFT MACB timestamps, Owner SID, .LNK references and the execution artifacts of the encryption utility all remain visible and correlatable.

Evidence-handling considerations

- **Capture volatile memory first.** If the desktop is live, RAM may hold plaintext fragments of encrypted documents, clipboard passphrases, BitLocker or EFS protectors, and mounted container keys — lost permanently at shutdown.
- **Preserve credential and key material.** Acquire NTUSER.DAT, DPAPI master keys under %APPDATA%\Microsoft\Protect\<SID>\, EFS certificates, browser Login Data, and pagefile.sys / hiberfil.sys, which frequently yield passwords or recovery keys.
- **Never decrypt on the original media.** Dictionary attacks, wordlist generation and EFS recovery run only on verified working copies so exhibit hashes stay untouched.
- **Enumerate streams and mismatches explicitly.** Configure the suite to list every secondary $DATA stream, flag hidden and system attributes, and pair magic-byte verification with Shannon entropy analysis (values above roughly 7.9 bits per byte suggest encryption or compression).

2.4 Evidence-Linkage Matrix (Question 11)

Question 11: Prepare an evidence-linkage table showing how file names, paths, timestamps, user accounts, metadata, and other artifacts could be correlated to a user.

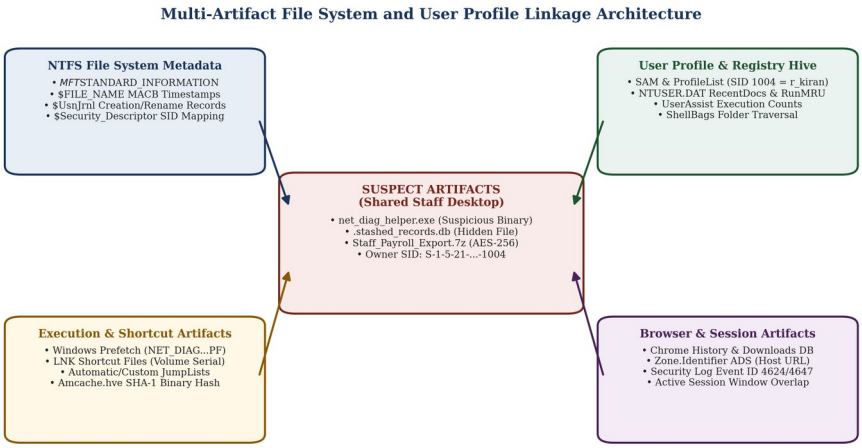


Figure 2.2: Multi-Artifact File-System, Registry, Execution and Browser Linkage Architecture

Table 2.3: Deliverable Evidence-Linkage Matrix Correlating Suspicious Files to Account r_kiran (Hypothetical)

Artifact and Path	MACB (IST, 11 Oct 2026)	Account / SID	File-System and Internal Metadata	Cross-Artifact Linkage
net_diag_helper.exe — C:\Users\Public\Documents\Diag	Born 14:12:04; Modified 14:12:05	Owner SID …-1004 (r_kiran)	$MFT record 64102; PE header hash; Zone.Identifier ADS with download URL	Chrome Downloads entry at 14:12:03 in r_kiran's profile; Prefetch NET_DIAG_HELPER.EXE-4A91B2.pf executed 14:13:19
.stashed_records.db (Hidden+System) — same Diag folder	Born 14:14:22; Modified 14:18:50	Owner SID …-1004 (r_kiran)	Flags +H +S; SQLite 3 header magic 53 51 4C 69 74 65	$UsnJrnl records creation and attribute change 85 s after the executable ran in that session
Staff_Payroll_Export_2026.7z — r_kiran\Documents\Archive	Born 14:22:10; Modified 14:22:41	Owner SID …-1004 (r_kiran)	Magic 37 7A BC AF 27 1C; entropy 7.98 bits/byte	RecentDocs and JumpList entry plus UserAssist showing 7zFM.exe launched 14:21:05
notes.txt:secret.cfg (ADS) — r_kiran\Desktop	$SI Modified 14:13:02; $FN Born 14:13:02	Owner SID …-1004 (r_kiran)	412-byte secondary $DATA stream on an 18-byte parent	ShellBags in UsrClass.dat confirm Desktop interaction at 14:13; stream holds the target subnet ranges
Staff_Payroll_Draft.xlsx.lnk — r_kiran\...\Recent	LNK born 14:19:15; target accessed 14:20:44	Profile owner r_kiran (…-1004)	Target $MFT entry 51908 (source since deleted); volume serial 7C94-A12B	Places the spreadsheet opening immediately before compression and encryption
Security.evtx logon record — winevt\Logs	Logon 14:05:12; logoff 14:31:09	r_kiran, Logon ID 0x3E7A41	Event 4624 Type 2 interactive; 4634 logoff	Only r_kiran held a console session during the whole artifact window

2.5 Conceptual ILookIX Acquisition and Analysis Workflow (Question 12)

Question 12: Explain a suitable workflow for importing/acquiring the forensic evidence and analyzing it using ILookIX, based on the concepts covered in the syllabus.

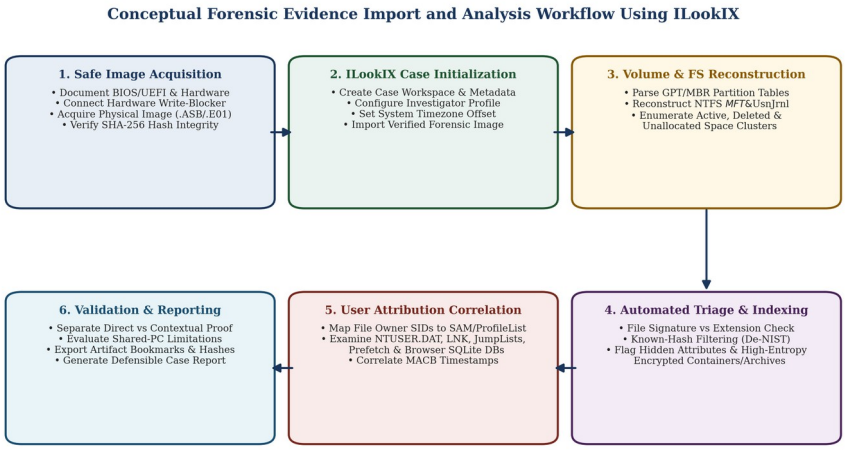


Figure 2.1: Conceptual Six-Stage Acquisition, Import and Analysis Workflow in ILookIX

- **Stage 1 — Write-protected acquisition and pre-import verification.** Image the NVMe SSD and SATA HDD through a hardware write-blocker with IXImager into .ASB or raw/E01 containers, then confirm the image hash matches the source SHA-256.
- **Stage 2 — Case workspace and read-only import.** On an isolated laboratory workstation, create case REC-DFIR-2026-02, record examiner details, the +42-second clock offset and the UTC+05:30 zone, and mount the verified image strictly read-only.
- **Stage 3 — Partition parsing and file-system reconstruction.** Parse the GPT layout, identify the NTFS volume, and rebuild the tree from $MFT, $Bitmap and $UsnJrnl so that deleted files, ADS entries, orphan files and unallocated clusters appear alongside live ones.
- **Stage 4 — Automated triage.** Run known-file hash filtering against good and bad sets to strip tens of thousands of benign system DLLs, verify each signature against its extension to flag mismatches and hidden attributes, and flag password-protected archives and encrypted Office documents.
- **Stage 5 — Multi-artifact account correlation.** Map Owner SID …-1004 to r_kiran through the SAM hive, then read that user's NTUSER.DAT, .LNK files, JumpLists, Prefetch entries and Chrome History and Downloads databases to confirm the session downloaded, executed and archived the files.
- **Stage 6 — Bookmarking, verification and export.** Bookmark findings by evidentiary category, re-check each at raw hex sector level to rule out parser error, and export paths, MACB values and hashes into the report.

2.6 Direct-Linking versus Supporting Artifacts (Question 13)

Question 13: Differentiate between an artifact that directly links a user to an activity and an artifact that merely provides supporting/contextual information.

Table 2.4: Comparative Evidentiary Weight of Direct and Contextual Artifacts

Dimension	Direct Account-Linking Artifact	Supporting / Contextual Artifact
Causal relationship	Produced inside a specific user's security context as an immediate byproduct of the act itself — download, execution, opening or encryption.	Produced system-wide or in shared space, or reflecting general settings; shows capability or chronology, not the actor.
Typical location	Inside the profile: NTUSER.DAT, AppData\Roaming\...\Recent\, the user's $Recycle.Bin\<SID>\ subfolder.	Shared or global: C:\Users\Public\, C:\Windows\Prefetch\, machine-level event logs.

Dimension	Direct Account-Linking Artifact	Supporting / Contextual Artifact
Examples here	r_kiran's .LNK and JumpList entry for the archive; UserAssist GUI launch of 7zFM.exe and net_diag_helper.exe; Chrome Downloads row in r_kiran's profile.	The executable sitting in C:\Users\Public\Documents\Diag\ reachable by all four accounts; the global Prefetch file proving only that the binary ran; the fact that 7-Zip is installed.
Attribution weight	High only when corroborated across several independent profile artifacts and an active Type 2 logon session.	Never sufficient alone; use to fix timeline consistency and system capability.

The practical test: if the artifact records a user action inside that user's own security context, it can link; if it records only that something happened on the machine, it corroborates.

2.7 Limitations and Alternative Explanations Before Attribution (Question 14)

Question 14: Identify at least three limitations or alternative explanations that investigators should consider before attributing the activity to a particular user.

Table 2.5: Five Counter-Explanations and the Corroboration Each Requires

Limitation / Alternative Explanation	Why It Matters	Corroboration Needed
Unattended unlocked session on a shared desktop	Artifacts bind actions to the authenticated SID, not to the person at the keyboard; a colleague could have used the open session between 14:12 and 14:25.	Office CCTV, biometric attendance logs, witness statements
Credential sharing or weak passwords	Staff commonly share logins for templates or printers, so another member could have logged in as r_kiran, possibly while that person was on leave.	Password-reset and shared-credential records; interview evidence
Administrator impersonation or cross-profile planting	admin_support with local admin rights can write into r_kiran's folders and run processes under that token using runas or PsExec.	Privileged-action logs, 4672/4673 events, service-creation records
Remote access or malware and scheduled-task execution	RDP, VNC or AnyDesk, or a trojan running in the session, inherits r_kiran's SID for the files it creates.	Logon Type in 4624, remote-connection and firewall logs, autoruns and persistence review
Timestamp forgery and clock drift	Attacker-controlled $STANDARD_INFORMATION times can make activity appear in another shift, and the CMOS clock ran 42 s fast.	Compare $SI against $FILE_NAME and monotonic $UsnJrnl sequence numbers; apply measured NTP offset

Accordingly, a report on this case should state that the artifacts are consistent with the activity occurring in r_kiran's interactive session and are not individually sufficient to conclude that r_kiran personally performed it.

3. CONCLUSION

Linking a suspicious file to a user on a shared machine is a layered exercise, not a single-artifact one. A complete pre-analysis inventory fixes what the machine could record; the storage, operating-system, file-system and metadata layers explain where the traces live; correct handling of hidden and high-entropy encrypted items preserves them; and a staged read-only examination workflow turns them into findings. On the hypothetical data above, concordant evidence — the Chrome download row, the $MFT Owner SID, UserAssist and JumpList execution records, the .LNK chain, and a console logon session spanning the artifact window — supports the conclusion that the executable was downloaded and the payroll archive created and encrypted within r_kiran's authenticated session

at FIN-DESK-04. It does not establish which individual typed at the keyboard, and the five counter-explanations above remain open until CCTV, attendance or privileged-action evidence is obtained. Because four staff members share the endpoint, examination also carries a privacy duty: review must be confined to the incident window so the uninvolved accounts' benign files are neither exposed nor reported.

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1. CASE OVERVIEW, FACTS AND ASSUMPTIONS

A laptop seized during a data-theft investigation contains an internal storage device and an attached external USB drive. Investigators must collect it without modifying the original media and carry out physical acquisition, forensic imaging, verification, carving where needed, and long-term safekeeping. IXImager and its ASB container are the designated tooling. This report answers Questions 15–22 and delivers the required acquisition flowchart, chain-of-custody form, step-by-step imaging procedure and evidence-integrity explanation. IXImager, ASB handling and hashing are described conceptually from syllabus principles; no tool was executed and every serial, date and hash reference below is hypothetical.

Table 3.1: Known Facts, Hypothetical Details and Working Assumptions

Category	Details
Known facts (assignment brief)	Laptop plus attached USB drive seized in a suspected data-theft case; collection must not modify originals; physical acquisition, imaging, carving and safekeeping are required; IXImager and ASB are in scope.
Hypothetical case details	A senior engineer in a VLSI and embedded research centre is suspected of exfiltrating circuit schematics (.pdf, .dwg) and firmware source archives (.zip) before joining a competitor. A Lenovo ThinkPad T14 Gen 3 with a 512 GB NVMe SSD is seized in Conference Room B with a 64 GB Kingston DataTraveler USB 3.2 still in its right-side port; the staging partition is believed formatted minutes earlier.
Working assumptions	Written institutional authorization with counsel oversight; the field procedure covers both powered-off and powered-on states, so a live RAM and encryption triage branch exists; carving on the USB is expected to succeed where SSD TRIM may have already zeroed blocks.

2. DETAILED ANSWERS TO THE ASSIGNMENT QUESTIONS (QUESTIONS 15–22)

2.1 Step-by-Step Collection and Transportation Before Imaging (Question 15)

Question 15: Prepare a step-by-step procedure for safely collecting and transporting the laptop and USB device before imaging.

Table 3.2: Eight-Step Field Procedure for Collection and Transportation

Step	Field Action	Purpose / Risk Avoided
1. Scene control and photography	Cordon the table so no one touches the keyboard or unplugs the drive; photograph screen state, indicator LEDs, power connection, network cables and the exact USB port used.	Prevents tampering and records the drive in situ, including which port held it.
2. Network isolation	Photograph then unplug the RJ-45; for Wi-Fi or Bluetooth use a Faraday bag or hardware radio kill switch.	Blocks remote wipe, lock and cloud-sync commands that could destroy evidence.

Step	Field Action	Purpose / Risk Avoided
3. Power-state triage	If powered off, do not boot — starting the OS rewrites timestamps and overwrites unallocated space. If powered on, check for active BitLocker, FileVault or VeraCrypt, capture RAM and any key material with a trusted live collector, and record OS time against a Stratum-1 UTC reference before shutdown.	Preserves volatile evidence and encryption keys; fixes the clock offset while the machine is still observable.
4. USB detachment and labelling	Wearing grounded anti-static gloves, remove the drive, record its markings and serial, cap the connector, and seal it alone in an anti-static Faraday bag (EV-302).	Avoids electrostatic damage and keeps the device's own timestamps uncontaminated.
5. Laptop inspection and bagging	Record make, model, serial and service tag (EV-301); confirm the optical drive is empty; label and bag the power adapter (EV-301A); place the machine in a padded anti-static tamper-evident bag.	Full exhibit description so nothing later proves missing or substituted.
6. Tamper-evident sealing	Seal with frangible tape; the seizing examiner writes initials, ID number and date across tape and bag.	Any opening visibly breaks the signature, proving integrity of storage.
7. Contemporaneous custody record	Complete the seizure log and chain-of-custody form on the spot with exact timestamps, location, exhibit descriptions and witness signatures.	Memory-based reconstruction later is challengeable; a field-signed entry is not.
8. Protected transport and intake	Carry sealed exhibits in a locked, shock-absorbing case in the climate-controlled cabin — never an uninsulated trunk exposed to heat, moisture, magnetic fields or vibration — and log intake into the access-controlled vault immediately.	Prevents media degradation and closes the custody gap between scene and laboratory.

2.2 Purpose of Physical Acquisition and Analysis on an Image (Question 16)

Question 16: Explain the purpose of physical acquisition and why analysis should normally be performed on a forensic image rather than the original storage media.

Physical acquisition produces a complete sector-by-sector bit-stream of the device at the raw block layer, independent of the operating system and file system. Every sector from LBA 0 to the end of the medium is captured, so deleted files, unallocated clusters, file slack, MBR or GPT structures, volume boot records, deleted partitions, shadow copies and Host Protected Areas are all preserved. Logical acquisition, by contrast, copies only files the file-system driver currently reports as allocated.

Table 3.3: Physical Bit-Stream Acquisition Compared with Logical Acquisition

Criterion	Physical Bit-Stream	Logical File / Partition
Scope	100% of addressable sectors, including deleted, unallocated and slack space plus boot structures.	Only active allocated files and folders visible to the OS.
Deleted data and carving	Full recovery of deleted $MFT records and carving of unallocated space.	No support for deleted-file recovery or carving.
Timestamp fidelity	Original $SI and $FN MACB values and raw sector offsets retained exactly.	Creation and access times often reset unless encapsulated in a logical container such as .L01.
When chosen	Mandatory where deletion, formatting or hidden data is suspected — as in this case.	Only where physical access is impossible, e.g. live cloud servers, locked phones, very large RAID arrays.

Analysis is then performed on the image rather than the exhibit for four reasons. Integrity: touching original media risks a write, an automount metadata update or operator error, whereas a read-only image leaves the exhibit sealed in the locker. Verifiability: an independent expert must be able to repeat the same tests on an identical state and obtain identical results, which is also the basis for certifying secondary electronic records under Section 63 of the Bharatiya Sakshya Adhiniyam, 2023. Hardware protection: a single imaging pass spares a failing head or a wear-limited flash controller from repeated read cycles. Throughput: images on laboratory storage can be indexed, carved and mounted by several tools at once, and a corrupted working copy can be regenerated from the master in minutes.

2.3 Forensic Imaging Workflow Using IXImager (Question 17)

Question 17: Design a forensic imaging workflow using IXImager. Include source identification, write protection, imaging, verification, hashing, and documentation.

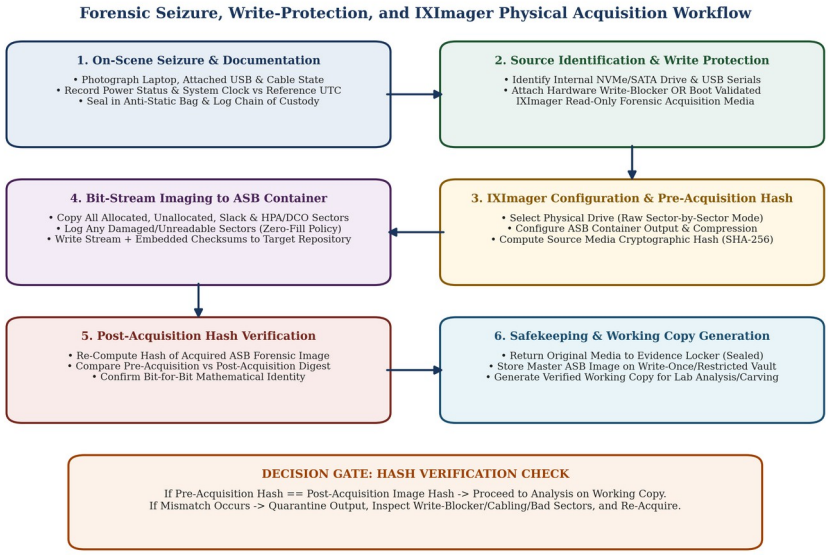


Figure 3.1: Deliverable Seizure, Hardware Write-Protection and IXImager Acquisition Workflow

Table 3.4: Six-Stage IXImager Imaging Workflow (Conceptual, Not an Execution Record)

Stage	Procedure	Verification Point
1. Source identification	Remove the internal NVMe drive (EV-301-SSD) or boot a validated IXImager forensic environment if the drive is soldered; prepare the USB drive (EV-302). Record make, model, firmware revision, capacity in bytes, LBA sector count and factory serial for both.	Device identity in the acquisition log matches the exhibit label.
2. Write protection	Connect each source only through a calibrated hardware write-blocker (T8u class for USB, T7u/T35u class for SATA/NVMe); in the bootable environment the kernel additionally marks block devices read-only (blockdev --setro).	Zero write operations reported by the blocker; dual-layer protection.

Stage	Procedure	Verification Point
3. Initialization and pre-hash	Select the target physical device (/dev/nvme0n1 or /dev/sdb), confirm the reported LBA count against the drive specification to expose any HPA or DCO, and run a full-device pre-acquisition SHA-256 and MD5 pass.	Baseline source digest exists before any byte is copied.
4. Bit-stream imaging into ASB	Image to a sanitized high-capacity repository drive in .ASB format with lossless compression and, if required, 2 GB or 4 GB chunk splitting, reading every sector in order; log any unreadable LBA range and pad the output block with zeros per standard operating procedure.	Sector count and bad-sector table recorded for the full span.
5. Post-acquisition verification	Re-read the completed ASB container, recompute SHA-256 and MD5 over the reconstructed bit-stream and compare with the pre-acquisition source hash.	A 100% digest match confirms bit-for-bit integrity.
6. Documentation and archival	Retain the automated log with UTC start and stop times, geometry, sector and bad-sector counts and digests; reseal the originals into the vault; archive one master ASB copy on write-once storage and pass a verified working copy to the analysis workstation.	Working copy hash re-verified before first mount.

2.4 Hash Values as Proof of Integrity Before and After Acquisition (Question 18)

Question 18: Explain how hash values can be used to demonstrate integrity before and after acquisition.

A cryptographic hash function maps an input of any length — here a bit-stream of roughly 10^9 sectors — to a fixed-length digest, 256 bits shown as 64 hexadecimal characters for SHA-256 as defined in FIPS 180-4, with MD5 retained for legacy comparison. Three properties carry the evidentiary weight. Determinism means the same sequence of sectors always yields the same digest. The avalanche effect means flipping a single bit out of trillions alters about half the output bits, producing an unrelated digest. Collision and pre-image resistance mean it is computationally infeasible to craft a substitute image with the same SHA-256 value.

Table 3.5: Three Integrity Checkpoints Across the Case

Checkpoint	Digest Computed Over	What It Proves
H_source — pre-acquisition	The write-blocked physical device at the bench, before imaging begins.	A baseline fingerprint of the exhibit as received.
H_image — post-acquisition	The reconstructed bit-stream of the finished ASB container.	The image is bit-for-bit identical to the source; nothing was added, lost or altered during acquisition.
H_working — before analysis and again before presentation	Each working copy at mount time and at report or hearing stage.	The data examined and reported on is still the data seized, with no modification in storage or handling.

When all three digests agree, the examiner offers mathematical rather than testimonial proof of continuity. Any mismatch is treated as a hard stop: the affected working copy is discarded and regenerated from the master, and the discrepancy is disclosed in the report.

2.5 Role of the ASB Container and Accompanying Information (Question 19)

Question 19: Describe the role of an ASB container in storing or managing forensic evidence and identify the information that should accompany the evidence.

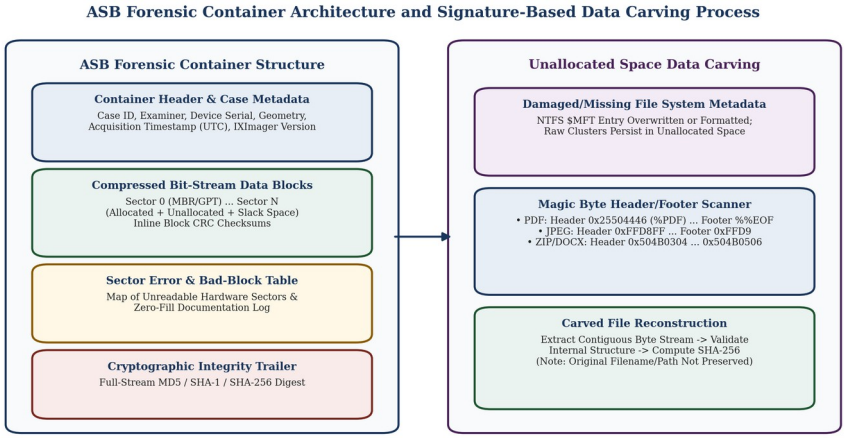


Figure 3.2: Internal Structure of an ASB Container (Left) and Signature-Based Carving Process (Right)

A raw flat image (.dd or .img) is an unindexed stream of sector bytes that can silently bit-rot or become separated from its external log. The ASB (Advanced Storage Block) container produced by IXImager for the ILookIX ecosystem instead acts as a self-validating wrapper, fulfilling four roles: it binds case and device metadata into the container header so an image can never become an unidentified orphan file; it applies lossless block compression, which pays off heavily across zeroed unallocated space; it embeds per-block cyclic redundancy checks plus full-stream digests in the trailer, so storage corruption is detected on read rather than discovered mid-analysis; and it mounts strictly read-only, preventing accidental modification during examination.

Table 3.6: Information That Must Accompany an ASB Evidence Container

Metadata Class	Required Fields
Case and administrative	Case identifier (REC-DFIR-2026-03), exhibit numbers (EV-301-SSD, EV-302-USB), investigating department, lead examiner name and ID, authorizing memo or warrant reference.
Source hardware	Device make, model, serial number, firmware revision, interface type (NVMe, SATA, USB), total byte capacity, total LBA count and sector size (512 or 4096 bytes).
Acquisition environment	Acquisition start and completion times in UTC, write-blocker make, model and serial, IXImager version, compression setting, and the documented BIOS/CMOS clock offset of the source laptop.
Integrity and errors	Pre- and post-acquisition MD5, SHA-1 and SHA-256 values, plus the complete bad-sector or unreadable-range map and the zero-padding applied.
Chain of custody	Every transfer of the container and the originals, with seal numbers, signatures and vault location.

2.6 Data Carving and When It Becomes Necessary (Question 20)

Question 20: Explain data carving and discuss when carving may be necessary if file-system information is unavailable or damaged.

Deleting a file or performing a quick format does not erase its contents: the directory entry or $MFT record is marked reusable and the clusters are marked unallocated in the $Bitmap or FAT, leaving raw bytes intact until overwritten. Carving becomes necessary precisely when the metadata that points to those bytes is missing or damaged — after a quick format of the USB drive; where old $MFT records have been reallocated to new files while the original clusters stay unallocated; where the partition table, superblock or boot sector is corrupt; or when examining raw unallocated swap and pagefile space. Standard parsing then finds nothing, so the examiner scans every unallocated sector sequentially, ignoring file-system structures, for known header signatures and footer or length markers, extracting each matching byte range into a standalone file and hashing it.

Table 3.7: Magic-Byte Signatures Used for Unallocated-Space Carving in an R&D Theft Case

File Type	Header Signature	Footer / Termination	Case Relevance
Adobe PDF (.pdf)	25 50 44 46 2D (%PDF-)	25 25 45 4F 46 (%%EOF)	Recovers deleted proprietary circuit schematic PDFs from USB unallocated clusters.
ZIP and Office Open XML (.zip, .docx, .xlsx)	50 4B 03 04 (PK..)	50 4B 05 06 (end of central directory)	Recovers firmware source archives and modern Office documents.
JPEG (.jpg)	FF D8 FF E0 / FF D8 FF E1	FF D9	Recovers photographs of whiteboards or prototype layouts.
7-Zip (.7z)	37 7A BC AF 27 1C	Payload length from header fields	Recovers deleted compressed or encrypted archives by header parsing.
AutoCAD drawing (.dwg)	41 43 31 30 (AC10..)	Internal structure or size cap	Recovers CAD designs when directory pointers are wiped.

Two limits must be reported alongside any carve. First, a carved file loses its external file-system identity — original name, folder path and $MFT MACB timestamps — though internal application metadata such as a PDF CreationDate and author persists inside the recovered object, so a carved file cannot by itself prove who handled it. Second, fragmented files split across non-contiguous clusters need bifragment gap carving or structure-aware carvers, and on the laptop's NVMe SSD, deletion followed by TRIM and controller garbage collection may already have zeroed the blocks; ordinary USB thumb drives rarely receive host-level TRIM, which is why the 64 GB drive is the realistic recovery target here.

2.7 Date/Time Problems and Documenting Uncertainty (Question 21)

Question 21: Discuss date/time problems in digital evidence and explain how investigators should document time-zone, clock, and timestamp uncertainties.

Table 3.8: Timestamp Problems and the Documentation Protocol for Each

Problem	Cause and Risk	Required Documentation
CMOS/BIOS clock drift or tampering	The hardware RTC may drift when it loses NTP, or a suspect may roll the clock back before copying files, shifting the whole timeline.	Read the RTC at BIOS/UEFI or during live triage, photograph it beside a Stratum-1 or GPS UTC reference, and record the signed delta $\Delta t = T_{BIOS} - T_{UTC}$ in the acquisition log.
UTC versus local storage	$MFT, event logs and registry LastWrite values are UTC; FAT32/exFAT directory entries hold the writing machine's local wall time with no zone tag.	Extract the SYSTEM TimeZoneInformation key (India Standard Time, bias −330, i.e. UTC+05:30) and state in every table whether a time is raw or converted.

Problem	Cause and Risk	Required Documentation
Resolution mismatch	NTFS resolves to 100 ns; FAT32 to 10 ms for creation, 2 s for modification, and whole days for last access.	Footnote the resolution limit per source; never read a FAT32 access stamp of 00:00:00 as midnight activity.
Copy and move semantics	A copied file receives a new creation time equal to the transfer moment while keeping the original modification time.	Record the created-greater-than-modified inversion as evidence of the copy event and its time.
Deliberate timestomping	Administrative utilities can rewrite $STANDARD_INFORMATION to arbitrary historical values.	Cross-compare $SI against kernel-written $FILE_NAME stamps and monotonic $UsnJrnl sequence numbers, and note any disagreement.

2.8 Chain-of-Custody Table for the Seized Laptop (Question 22)

Question 22: Prepare a chain-of-custody table for the seized laptop from collection through laboratory examination.

Table 3.9: Deliverable Chain-of-Custody Log, Conference Room B to Laboratory Examination (Hypothetical)

Date / Time (IST)	Exhibit	Released By	Received By	Route and Purpose	Seal / Hash Verification
13 Oct, 11:15	EV-301 laptop; EV-302 USB	Scene security officer (R. Venkataraman)	Lead field examiner	Conference Room B to field transport case — seizure after photography and safe power-down.	Sealed in anti-static bags #REC-301, #REC-302; tape initialed and dated
13 Oct, 11:50	EV-301; EV-302	Lead field examiner	Evidence vault custodian (M. Karthikeyan)	Field transport to DFIR vault Locker #C4 — secure storage.	Seals verified intact; entered in intake register
13 Oct, 14:00	EV-301; EV-302	Evidence vault custodian	Forensic imaging specialist	Vault to imaging bench #2 — write-blocked physical acquisition via IXImager.	Seal #REC-301 broken and recorded; new seal applied after imaging
13 Oct, 14:15–16:40	Sources to master ASB images	Forensic imaging specialist	Forensic imaging specialist	Bench acquisition; images written to repository; H_source and H_image computed.	SHA-256 of source equals SHA-256 of image; bad-sector table logged
13 Oct, 17:00	EV-301; EV-302	Forensic imaging specialist	Evidence vault custodian	Bench to Locker #C4 — original media returned to secure storage.	Resealed in new signed bags #REC-301B, #REC-302B
14 Oct, 09:30	Working copies EV301_work.asb, EV302_work.asb	Forensic imaging specialist	Forensic analyst	Lab storage to analysis workstation — read-only mounting, file-system analysis and carving.	Working-copy SHA-256 verified before first mount

3. CONCLUSION

The methodology above yields a defensible path from conference-room seizure to courtroom-ready examination: disciplined field handling, radio isolation and power-state triage; hardware write-blocking; full bit-stream imaging through IXImager into self-checksumming ASB containers; pre- and post-acquisition SHA-256 verification; signature carving of the USB drive's unallocated space, which is the realistic recovery target given TRIM exposure on the SSD; explicit documentation of clock

offset, time-zone and resolution uncertainty; and an unbroken custody register covering both the physical exhibits and the derived images. Two limitations follow directly from the physics rather than the procedure — carved files carry no file-system identity, so a recovered .pdf proves content existed, not who copied it, and SSD TRIM may have destroyed data that on the USB drive would have survived. Reporting both preserves the objectivity that Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 presumes of the certifying examiner.

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1. CASE OVERVIEW, FACTS AND ASSUMPTIONS

An employee is suspected of deleting documents and browser artifacts after unauthorized activity. A verified forensic image of the Windows workstation exists and Volume Shadow Copies are present on the imaged volume, so the team must select relevant evidence, discard irrelevant data, analyze using event, cloud, lead and shadow-copy examination, validate what it recovers, and reconstruct the sequence of activity. This report answers Questions 23–30 and delivers the four required outputs: an evidence classification table, an event timeline, a Volume Shadow Copy recovery workflow, and a validation checklist. No tool was executed; all identifiers, hashes and timestamps are hypothetical constructs used to illustrate method.

Table 4.1: Known Facts, Hypothetical Details and Working Assumptions

Category	Details
Known facts (assignment brief)	Suspected deletion of documents and browser artifacts after unauthorized activity; a verified image of a Windows system is available; Volume Shadow Copies exist on the imaged volume; selection, reduction, analysis, validation and timeline reconstruction are required.
Hypothetical case details	Procurement officer emp_naveen (SID …-1108) allegedly altered vendor evaluation scores, uploaded Bid_Pricing_Matrix_2026.xlsx and Vendor_Margin_Notes.docx to a personal cloud dropbox between 9 and 11 Oct 2026, then deleted the files and cleared Edge and Chrome history on 11 Oct. The 512 GB Windows 11 NTFS volume was imaged on 12 Oct. The live folders are empty and the live History database holds no entries before 12 Oct, but System Volume Information contains two intact shadow copies: ShadowCopy1 of 9 Oct 18:00 IST and ShadowCopy2 of 10 Oct 20:00 IST.
Working assumptions	Every filename, snapshot GUID, timestamp and hash is illustrative; the suspect held no local administrator rights and did not run vssadmin delete shadows /all, which is precisely why the historical copies survived.

2. DETAILED ANSWERS TO THE ASSIGNMENT QUESTIONS (QUESTIONS 23–30)

2.1 Selecting and Categorizing Evidence Before Detailed Analysis (Question 23)

Question 23: Explain how investigators should select and categorize digital evidence before detailed analysis.

A 512 GB Windows image holds hundreds of thousands of files and millions of registry and log records, so manual review of everything is both impossible and error-prone. Selection proceeds in five stages.

- **1. Filter out known-benign system files.** Hash every allocated file and compare against the NIST NSRL Reference Data Set, removing stock DLLs, fonts, drivers and signed binaries — typically 75 to 85 percent of the file set — so attention lands on user-created, modified, deleted and suspicious content.
- **2. Screen for known-bad and mismatched items.** Check survivors against threat and anti-forensic tool hash sets (wipers, cleaner utilities such as BleachBit, exfiltration helpers) and flag files whose magic bytes contradict their extension.

- **3. Bound the review by time.** Filter $MFT, $UsnJrnl and event-log records to the 9 to 12 October 2026 window so creation, modification, access and deletion events inside the incident period are prioritized.
- **4. Isolate artifact domains.** Split the workspace into user profile directories, browser profile folders, deletion-tracking structures ($Recycle.Bin, $UsnJrnl:$J, $LogFile), execution and linkage artifacts (NTUSER.DAT, .LNK, JumpLists, Prefetch), and the historical shadow copies.
- **5. Categorize formally before correlation.** Tag every retained artifact into the five-category scheme of Question 24, so primary user content, system telemetry, suspicious tools, recovered deletions and contextual evidence can be correlated without mixing evidentiary classes.

2.2 Five-Category Evidence Classification Scheme (Question 24)

Question 24: Prepare a classification scheme for user-created files, system/application files, potentially suspicious files, deleted/recovered artifacts, and supporting/contextual evidence.

Table 4.2: Deliverable Five-Category Digital Evidence Classification Scheme

Category	Defining Criteria and Case Artifacts	Role / Value	Examination Method
1. User-created files	Content authored or edited in profile folders: Bid_Pricing_Matrix_2026.xlsx, Vendor_Margin_Notes.docx, Evaluation_Summary_Draft.pdf.	Primary target objects; internal metadata (author, last saved by, revision) shows what business data existed.	Compare live folder state with each shadow copy; compute SHA-256; inspect OOXML docProps/core.xml.
2. System and application files	Telemetry and journals: $MFT, $UsnJrnl:$J, Security and System .evtx, registry hives, browser SQLite databases, Prefetch.	Objective behavioural ledger showing when files were created, edited, deleted and which programs ran.	Parse with read-only tools; validate against raw hex structures.
3. Potentially suspicious files	Wipers or portable cleaners (bleach_portable.exe, purge_hist.ps1), encrypted staging archive tender_pack.zip, PowerShell script-block logs.	Evidence of intent, anti-forensic capability or staging before deletion.	Hash and signature screening; Prefetch and Amcache execution times; PowerShell Event ID 4104.
4. Deleted and recovered artifacts	Version 1 and 2 of the spreadsheet recovered from ShadowCopy1 and 2; $I49A21.xlsx in $Recycle.Bin; URL visits recovered from the pre-wipe History database.	Directly reconstructs what the user tried to purge — the core evidentiary payload of this case.	Mount snapshots read-only; extract intact files; parse SQLite freelist and WAL; validate structure and hash.
5. Supporting and contextual evidence	SAM and ProfileList mapping SID …-1108 to emp_naveen; TimeZoneInformation key; the spreadsheet .LNK and Excel JumpList f01b4d95….automaticDestinations-ms.	Corroborates that deleted files existed at stated paths and were opened in a specific profile session.	Match .LNK target $MFT record and sequence numbers against recovered shadow-copy records.

2.3 Event Analysis and Sequence Reconstruction (Question 25)

Question 25: Explain how Event Analysis can assist in reconstructing a sequence of activities.

Event Analysis treats every timestamped artifact — a logon record, an $MFT timestamp change, a $UsnJrnl reason flag, a registry update, a browser visit — as an atomic event, then orders events on a single normalized time axis so cause-and-effect chains emerge. Applied to a case where files were deleted and history cleared, it contributes in four ways.

- **Establishes causal sequence.** Logon (4624) → document creation ($MFT Born) → editing (JumpList plus $UsnJrnl DATA_OVERWRITE) → browser launch and cloud upload (URL visit preserved in a shadow copy) → deletion ($UsnJrnl FILE_DELETE plus $I record in $Recycle.Bin) → history clearing (History table truncation) → logoff (4634).
- **Exposes gaps and tampering.** A live browser database showing zero history during a two-hour session in which $UsnJrnl and Prefetch record heavy browser and file activity is an anomalous void, and it directs the examiner to the shadow copies taken before the wipe.
- **Clusters micro-events into actions.** One save or one Recycle Bin emptying generates dozens of low-level file-system and registry events within milliseconds; grouping them into comprehensible human actions is what makes the timeline readable to a non-technical reader.
- **Cross-validates order with counters.** Pairing timestamps with strictly increasing NTFS update sequence numbers and Event Log EventRecordID values fixes the true order of activity even where the system clock was moved.

2.4 Purpose of Cloud Analysis and Lead Analysis (Question 26)

Question 26: Explain the purpose of Cloud Analysis and Lead Analysis in an investigation and identify the kinds of questions each may help answer.

Table 4.3: Cloud Analysis Compared with Lead Analysis

Dimension	Cloud Analysis	Lead Analysis
Core purpose	Identify and correlate the local artifacts left by cloud sync clients and web cloud sessions to establish what data moved between the endpoint and remote repositories.	Extract actionable pointers — keyword hits, addresses, external URLs, device serials, collaborator names, hashes — that route the examiner to evidence sources not yet examined.
Primary artifacts	SyncEngineDatabase.db, filecache.dbx, Google Drive metadata stores, OneDrive folders, OAuth tokens, cookies, and REST upload URLs recovered from shadow copies.	WordWheelQuery and omnibox searches, document Author, Comments and hyperlink metadata, $UsnJrnl names of files no longer on disk, network and removable-drive paths inside .LNK files.
Questions it answers	Were the deleted documents uploaded or synced before deletion? Which cloud account was authenticated? Do sync logs carry the file hash, object ID and sharing permissions? Does an intact copy survive on the provider or in cloud trash?	Who else collaborated on or received the file? What vendor names or codenames was the user searching for? Did the file also go to a removable drive or network share before local deletion? Did the user search for ways to permanently delete files or clear shadow copies?

The two are complementary rather than alternative: leads generated from local artifacts tell the examiner which cloud artifacts to request, and cloud findings — an upload timestamp and object hash — close the loop by corroborating or contradicting the local timeline.

2.5 Volume Shadow Copy Analysis for Recovery and Comparison (Question 27)

Question 27: Describe how Volume Shadow Copy analysis can help recover or compare earlier versions of evidence.

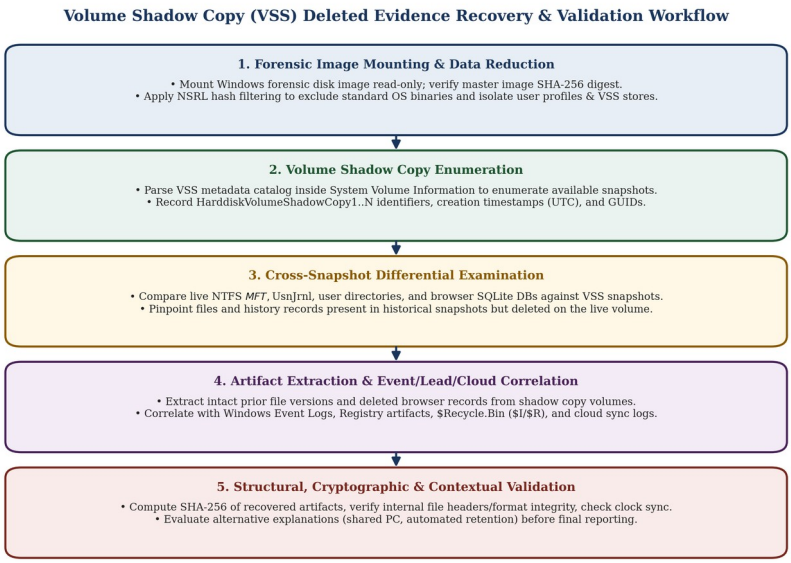


Figure 4.1: Deliverable Volume Shadow Copy Recovery and Differential Validation Workflow

VSS is a copy-on-write mechanism in volsnap.sys: when a block on the live volume changes after a snapshot, the original 16 KB block is first written into the shadow storage under System Volume Information. Each shadow copy therefore carries its own self-consistent NTFS state — its own $MFT, directory trees, profile folders, registry hives and browser databases — at the instant it was taken, which is what makes the following four analyses possible.

- **Full recovery of purged files, bypassing $Recycle.Bin and TRIM.** A Shift+Delete on 11 October never places the spreadsheet in the Recycle Bin, yet ShadowCopy1 and ShadowCopy2 retain its allocated blocks and $MFT record, so the file comes back intact with original name, path and timestamps.
- **Differential version comparison.** Extracting both versions and diffing them cell by cell proves not merely that the file existed but which vendor pricing cells changed between 9 and 10 October, before deletion on 11 October.
- **Recovery of cleared browser databases and hives.** Mounting ShadowCopy2, taken before the clearing event, yields the Edge History database as it stood on 10 October at 20:00 IST, exposing the exact cloud upload URLs visited.
- **Historical $MFT and $UsnJrnl comparison.** Differencing the $MFT inside each snapshot against the live $MFT and $UsnJrnl brackets the interval in which each file existed and the moment it disappeared.

2.6 Hypothetical Lifecycle Timeline (Question 28)

Question 28: Construct a hypothetical timeline showing how a file could be created, modified, copied, deleted, and later recovered from a Volume Shadow Copy.

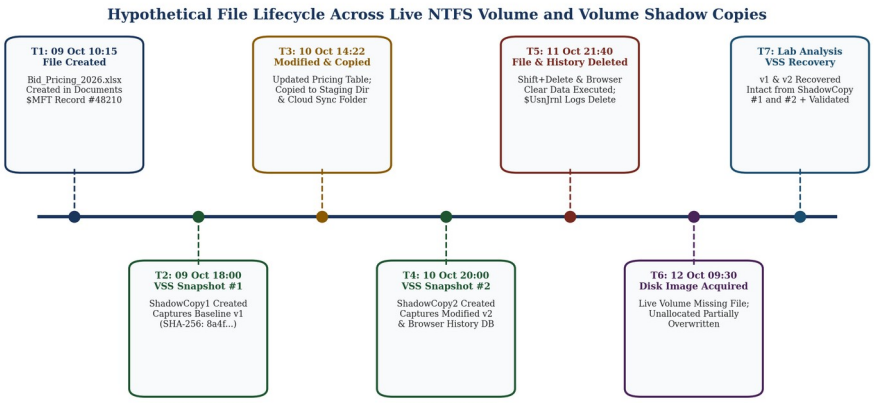


Figure 4.2: Hypothetical Lifecycle — Creation, Modification, Copying, Deletion and Shadow Copy Recovery

Table 4.4: Deliverable Hypothetical Event Timeline for Bid_Pricing_Matrix_2026.xlsx

Timestamp (IST)	Stage	Volume / Artifact Location	Technical State and Significance
9 Oct, 10:15:22	1. Creation	Live C: \Users\emp_naveen\Documents\	$MFT record #48210 seq 3; $SI and $FN born 10:15:22; version 1 SHA-256 8a4f19c2…
9 Oct, 18:00:04	2. Snapshot 1	System Volume Information \ ShadowCopy1	Scheduled restore point freezes an immutable block-level copy of version 1.
10 Oct, 14:22:18	3. Modification	Live volume plus Excel JumpList	Vendor bid rows edited and saved; $SI modified updates; version 2 SHA-256 3c91e07b…
10 Oct, 14:35:09	4. Copy and upload	Downloads\Stage\ plus Edge History	Copy created 14:35:09 with the older modified time 14:22:18 — the classic copy inversion — then uploaded to personal cloud.
10 Oct, 20:00:11	5. Snapshot 2	ShadowCopy2	Windows Update checkpoint preserves version 2, the staged copy and an intact Edge History database.
11 Oct, 21:40:55	6. Deletion and wipe	Live $UsnJrnl:$J and browser profile	Shift+Delete on both copies logged as FILE_DELETE; browsing history cleared and the History table emptied.
12 Oct, 11:20:00	7. Recovery	Read-only mounted image in the laboratory	Examiner mounts both snapshots and recovers version 1, version 2 and the pre-wipe History database.

2.7 Validating a Recovered Artifact (Question 29)

Question 29: Explain how the recovered artifact should be validated before it is included as a finding in the final report.

A recovered file must never enter a report on the strength of one tool's summary line. The following checklist is applied in order, and a finding is reported only when every step passes.

Table 4.5: Deliverable Pre-Reporting Validation Checklist for Recovered VSS Artifacts

Step	Verification Procedure	Acceptance Criterion
VAL-01 Container integrity	Confirm the mounted image's SHA-256 matches the acquisition hash; verify snapshot catalog GUID and creation time.	Image hash verified; snapshot GUID and creation timestamp intact.
VAL-02 Structural integrity	Check magic bytes (50 4B 03 04 for OOXML, 53 51 4C 69 74 65 for SQLite) and run an integrity check on recovered databases.	File opens without repair; SQLite integrity check returns ok.
VAL-03 Dual-tool cross-check	Recover and parse with two independent toolchains and compare digests and timestamps.	Bit-for-bit identical files and identical $MFT timestamps from both.
VAL-04 Live-volume corroboration	Correlate $MFT record #48210 seq 3 with surviving $UsnJrnl deletion record, .LNK target metadata and JumpList entries.	Live artifacts reference the same record number recovered from the snapshot.
VAL-05 Internal metadata	Inspect OOXML core properties for creator, last modified by, created and modified times and compare with NTFS stamps.	Internal and file-system times agree within one second; no timestomping indicated.
VAL-06 Temporal and attribution boundary	Apply the measured CMOS offset, confirm UTC to IST normalization, and state explicitly what the artifact proves.	All times normalized; existence and deletion proven, identity not asserted.

2.8 Sources of Error and Alternative Explanations (Question 30)

Question 30: Identify possible sources of error or alternative explanations that could affect the interpretation of timestamps and deleted artifacts.

- **Recovery from a shadow copy does not prove who deleted the file.** Finding the spreadsheet in ShadowCopy2 and noting its absence on the live volume establishes only that it existed at 20:00 IST on 10 October and was removed before imaging on 12 October. Attribution of the deletion additionally requires the live $UsnJrnl FILE_DELETE record, the SID subfolder under $Recycle.Bin and a correlated logon session — and even those speak to account, not person.
- **Snapshot time is not file time.** The creation time of ShadowCopy1 (18:00) is not the modification time of the file inside it (10:15). A snapshot preserves state as at capture, so activity that occurred and was deleted between two snapshots can appear in neither, creating a false sense of a quiet interval.
- **Automated or policy-driven deletion.** Storage Sense, enterprise cleanup scripts, a browser set to clear cookies on exit, antivirus quarantine, or cloud Files On-Demand dehydration can remove local files or truncate databases with no user intent at all.
- **Copy semantics and clock skew.** Moving or copying across folders and volumes rewrites $STANDARD_INFORMATION while leaving $FILE_NAME untouched, and an unsynchronized RTC shifts every derived time — both can make innocent file handling resemble staged exfiltration, or hide the real moment.
- **Tool parsing error and shadow-copy access method.** Snapshot mounting through the live operating system can alter state, and SQLite freelist or WAL remnants may recombine rows incorrectly, so recovered content must be verified structurally and by a second toolchain before it is quoted.

3. CONCLUSION

Volume Shadow Copies convert a deletion that looked like destruction into a recoverable, comparable and datable event. In this hypothetical case, systematic selection and NSRL reduction narrowed a 512 GB image to a tractable artifact set; the five-category scheme kept user content, telemetry, suspicious tools, recovered deletions and context from being conflated; event analysis supplied causality that no single artifact could; shadow-copy examination recovered both versions of the pricing spreadsheet and the pre-wipe browser history; and the six-step validation checklist certified those recoveries before they were reported. The defensible finding is therefore that Bid_Pricing_Matrix_2026.xlsx existed, was edited between 9 and 10 October, was copied to a staging folder and uploaded through the browser, was deleted along with browsing history on 11 October, and was recoverable in intact form from ShadowCopy1 and ShadowCopy2 — all within emp_naveen's profile. What remains unresolved is who performed the deletion and whether the cloud copy still exists at the provider, since a snapshot preserves state but records no actor.

4. REFERENCES

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1. CASE OVERVIEW, FACTS AND ASSUMPTIONS

A student account is suspected of unauthorized access to an online service. Investigators hold authorized access to six distributed evidence domains — a Windows computer, browser artifacts, email, messaging records, a mobile phone, and Internet or cloud records — and must correlate across them while reckoning with the technical and legal difficulties that multi-source forensics creates. This report answers Questions 31–38 and delivers the multi-source evidence map, the correlation matrix, the challenge analysis and the final investigative summary. No tool was executed; all addresses, tokens, hashes and timestamps are hypothetical, with documentation address space (RFC 5737) used for external hosts.

Table 5.1: Known Facts, Hypothetical Details and Working Assumptions

Category	Details
Known facts (assignment brief)	A student account suspected of use in a cyber incident involving unauthorized access to an online service; six authorized evidence domains; the task is cross-source correlation under legal and technical constraints.
Hypothetical case details	On 13 Oct 2026, 19:44–19:52 IST (14:14–14:22 UTC), the college repository portal (portal.rec-cloud-edu.example) logged an API bulk download of restricted grant proposals under account stu_aravind. The student states he was doing a programming assignment on his Windows 11 laptop in the hostel and had ten minutes earlier clicked a 'Library Overdue Verification' link delivered by email and by WhatsApp or Telegram on his Android phone. The competing hypotheses are deliberate misuse and a cross-device OAuth session-token phishing attack.
Working assumptions	All examinations proceed under formal institutional authorization with documented consent for the personal laptop and phone; IP addresses such as 198.51.100.74 (phishing host) and 203.0.113.88 (external VPS) are RFC 5737 documentation values; no live tool execution is claimed.

2. DETAILED ANSWERS TO THE ASSIGNMENT QUESTIONS (QUESTIONS 31–38)

2.1 Evidence from Windows and Other Operating Systems (Question 31)

Question 31: Identify the potential evidence available from Windows and other operating systems in this case.

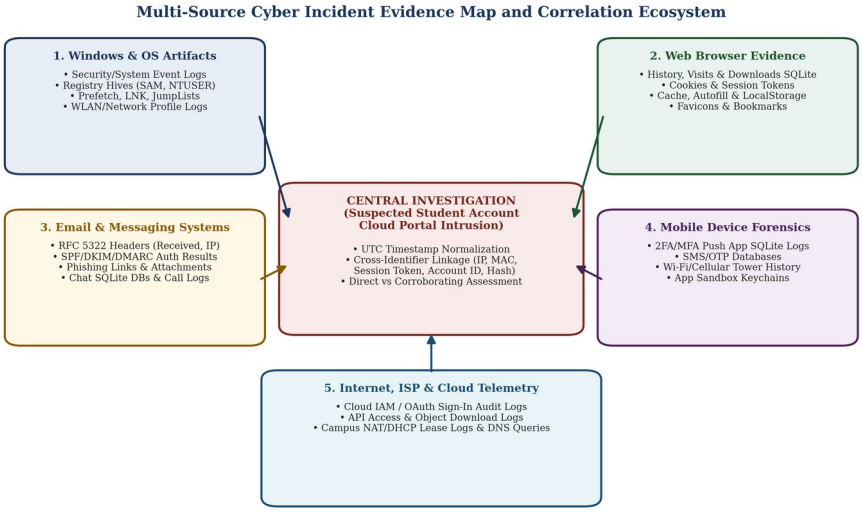


Figure 5.1: Deliverable Multi-Source Evidence Map Across the Six Authorized Domains

Table 5.2: Comparative Operating System Evidence Matrix (Windows, Linux, macOS)

Evidence Category	Windows Artifacts	Linux / macOS Artifacts	Value in This Case
Authentication and session logs	Security.evtx — 4624 logon, 4634/4647 logoff, 4625 failure, 4672 special privileges; TerminalServices-LocalSessionManager for RDP.	/var/log/auth.log or secure, wtmp, btmp, lastlog, journald; macOS Unified Log and /var/log/asl.	Shows whether the student was interactively logged into the laptop throughout the 19:44–19:52 IST window, and whether any remote login occurred.
Network and wireless telemetry	WLAN-AutoConfig.evtx (SSID, BSSID, DHCP lease); registry NetworkList\Profiles and Interfaces; SRUM (SRUDB.dat) per-application byte counters.	NetworkManager and wpa_supplicant syslog entries, system-connections files; macOS com.apple.airport.preferences .plist.	Fixes the campus address 172.16.8.112 held by the laptop and quantifies what each application actually transmitted.
Process execution and command history	Prefetch *.pf, Amcache.hve, Shimcache, UserAssist; PowerShell ConsoleHost_history.txt and Script Block Logging (4104).	bash/zsh history, acct or auditd EXECVE records; macOS history, LaunchAgents, TCC.db.	Tests whether any scraper (curl, wget, python-requests) or remote-access tool ran on the endpoint at all.
File system and shortcut artifacts	NTFS $MFT, $UsnJrnl:$J, $LogFile, $Recycle.Bin; .LNK, JumpLists, ShellBags.	ext4/XFS inode crtime and mtime, recently-used.xbel, Trash info files; APFS FSEvents, Spotlight store, .DS_Store, QuarantineEventsV2.	Establishes whether the grant PDFs were ever downloaded, opened or deleted locally — absence here is itself probative.

2.2 What Browser Artifacts Reveal and Why Care Is Needed (Question 32)

Question 32: Explain what browser artifacts may reveal about user activity and why browser evidence must be interpreted carefully.

Browsers keep relational SQLite stores and cache trees inside the profile (for example AppData\Local\Google\Chrome\User Data\Default), which can reveal four classes of activity: navigation history with page titles, visit counts, referring URLs and transition type (typed, link-clicked, redirected, reloaded); the downloads table with target path, source and referrer URL, byte count and interrupt reason; cookies, local and session storage holding authentication cookies and OAuth session identifiers; and cache, favicons, autofill and Web Data preserving response fragments, search terms and saved-credential metadata.

Reading a URL row as proof that a human viewed that page is a classic error, for five reasons. Browsers DNS-prefetch, pre-render and load iframes, analytics beacons and extension traffic, generating history entries with no user action. Redirect chains — inspect the CORE and QUALIFIER masks such as IS_REDIRECT_MASK and SERVER_REDIRECT — can carry a user through a malicious domain in milliseconds after an innocuous click. Profile sync replicates history, bookmarks and open tabs between devices, so entries created remotely can appear in the local database; check synced flags and originator_cache_guid. Session restore refreshes background tabs and rewrites timestamps. Finally, incognito activity lives mostly in RAM and vanishes at window close, leaving only fragments in DNS cache, pagefile or favicons.

Correlating browser rows with independent sources — network flow records, file-system artifacts, mail headers — is therefore mandatory before any browser entry is quoted as a finding.

2.3 Email and Messaging Evidence and Its Correlation (Question 33)

Question 33: List the types of evidence that may be available from email and messaging systems and explain how they can be correlated with other artifacts.

- **Headers and authentication telemetry.** RFC 5322 headers give the bottom-to-top chain of Received hops with mail-transfer hostnames, relaying IP addresses and offset-stamped times, plus Message-ID, Return-Path, Reply-To, X-Originating-IP and Authentication-Results recording SPF, DKIM and DMARC pass or fail.
- **Body content, links and attachments.** The lure text, disguised anchor targets, attachment filenames, MIME types and SHA-256 digests are the pivot points into every other source.
- **Messaging databases and media caches.** Platforms such as WhatsApp (msgstore.db or wa.db), Telegram caches and Teams IndexedDB hold sender and recipient identifiers, message text, shared URLs, attachment metadata, read receipts and call logs.

Correlation follows four pivots: match the phishing URL in the message body against the identical URL and time in the mobile or desktop browser history; match Message-ID and delivery time against campus mail gateway logs; match an attachment hash against a file in the Downloads folder and its NTFS Zone.Identifier stream; and match the originating IP in the headers against the address that later authenticated to the cloud portal — in this scenario, evidence that the lure and the intrusion came from different hosts.

2.4 Mobile Phone Considerations and Challenges (Question 34)

Question 34: Explain the forensic considerations when examining a mobile phone and identify challenges associated with mobile evidence.

- **Isolate radios immediately.** A connected phone can receive an over-the-air wipe from Find My Device or Find My, or be overwritten by incoming traffic; set airplane mode with screen actions photographed and store the device in a shielded Faraday bag.
- **Preserve the unlock state.** Android file-based encryption and iOS data protection distinguish Before First Unlock, where keys stay wrapped in hardware, from After First Unlock, where they sit in volatile memory. A powered-on AFU phone must be kept charged inside the enclosure so it does not revert to BFU and become unreadable.
- **Select the acquisition level deliberately.** Logical backup APIs, file-system extraction of application sandboxes and SQLite stores, or physical acquisition where the chipset allows it — documenting the method and hashing each output.
- **Target the high-value artifacts.** Authenticator and push-approval logs, SMS one-time-password databases, mobile browser history, messaging stores, and Wi-Fi or cellular location history decide account-compromise cases like this one.

The challenges that follow are structural, not incidental: hardware-backed encryption with rate limiting in a Secure Enclave or Titan-class chip makes brute force unrealistic without a passcode; application sandboxing plus SQLCipher and disappearing-message timers keeps even acquired data opaque; internal UFS or NVMe flash cannot simply be attached to a SATA write-blocker; and frequent operating-system and application updates break parser schemas between examinations.

2.5 Internet and Cloud Evidence Sources (Question 35)

Question 35: Discuss Internet and cloud-based evidence sources that may be relevant to the investigation.

Table 5.3: Five Cloud and Internet Evidence Sources and Their Evidentiary Contribution

Source	What It Records	Contribution
Identity provider and SSO logs	Entra ID, Workspace admin audit or Okta sign-in records: UTC timestamp, principal name, source public IP, ASN and geolocation, user agent, MFA method, OAuth client and token identifier.	The authoritative answer to who authenticated, from where, with what client, and whether MFA was satisfied.
Application and storage audit logs	Portal web-server access logs, CloudTrail, Azure Monitor or Cloud Audit logs, database query logs: exact endpoints called, status codes, bytes transferred, objects accessed.	Converts 'account was used' into 'these 14 objects were exported at these times'.
OAuth grant and consent trails	Token issuance records and consent grants, including which external address redeemed a refresh or access token.	The decisive trail in adversary-in-the-middle cases, where a legitimate login hands a token to an attacker.
Campus gateway, NAT, DHCP, DNS	Firewall and NAT translations, DHCP leases, 802.1X RADIUS wireless authentication, recursive DNS query logs.	Binds a shared public address to an internal host and MAC, or shows the absence of any such flow.
External registry and threat records	WHOIS or RDAP registration, passive DNS history, hosting abuse records and threat-intelligence reports.	Identifies the infrastructure behind a phishing host, subject to legal process outside the institution.

2.6 Cross-Source Correlation Matrix (Question 36)

Question 36: Prepare a cross-source correlation table linking at least five hypothetical artifacts from the computer, browser, email/messaging, mobile, and cloud sources.

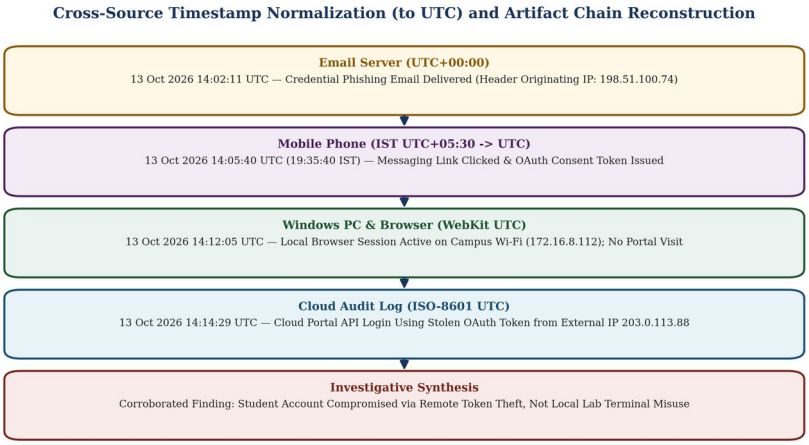


Figure 5.2: Cross-Epoch Timestamp Normalization to UTC and Artifact Chain Reconstruction

Table 5.4: Deliverable Cross-Source Correlation Matrix (Six Hypothetical Artifacts)

Artifact / Domain	Record Examined	Time (UTC)	Pivots	Analytical Significance
ART-501 Email and messaging	Webmail .eml headers plus Android msgstore.db	14:02:11 (mail); 14:03:05 (chat)	Message-ID 99a1@mail-verify.example; sender 198.51.100.74; lure URL rec-portal-auth.example	Delivery of a spear-phishing lure to stu_aravind, SPF softfail on a spoofed SSO link.
ART-502 Mobile phone	Chrome mobile history and Microsoft Authenticator log	14:05:40 (click); 14:06:12 (MFA approved)	Pixel 7a, cellular IP 49.37.210.18; request req-77b2; token tok_9f3c	The student clicked the link on cellular and approved the push prompt the attacker's proxy relayed — the token is born here.
ART-503 Windows computer	Security.evtx 4624, WLAN-AutoConfig, $MFT and $UsnJrnl	13:30–15:00 continuous session	Host ARAVIND-LAPTOP; 172.16.8.112; NAT 14.139.180.12	Local work on C++ coursework; no grant PDF in $MFT, journal or unallocated space.
ART-504 Desktop browser	Chrome History, Downloads and Cookies	14:12:05–14:25:19	Visited docs.python.org, github.com; zero portal visits	The laptop browser never touched the repository portal and downloaded nothing.
ART-505 Cloud service	IAM sign-in log and REST API audit trail	14:14:29 (login); 14:15:02–14:21:48 (export)	Account stu_aravind; source 203.0.113.88; UA python-requests/2.31.0; token tok_9f3c	The export ran on an external Linux host replaying the token minted at ART-502 — not from the laptop or the phone.
ART-506 Campus network	Firewall NetFlow and recursive DNS logs	14:00:00–14:30:00	Client 172.16.8.112, MAC A4-B1-C2-…; zero flows to the portal	Infrastructure-layer confirmation that the student laptop generated no traffic to the compromised portal.

The chain closes because the same token string, tok_9f3c, appears at 14:06:12 UTC on the phone and at 14:14:29 UTC in the cloud log from a different network and a different user agent — a two-minute hand-off from a human-approved prompt to an automated exporter.

2.7 Challenges in Distributed Multi-Device and Cloud Forensics (Question 37)

Question 37: Explain at least six major challenges in digital forensics when evidence is distributed across multiple devices and cloud services.

Table 5.5: Seven Major Challenges and Their Mitigations

Challenge	Technical and Evidentiary Impact	Mitigation
Diverse epochs, clock skew	Windows FILETIME counts 100 ns intervals from 1601, Android and Linux use Unix seconds, Chromium uses WebKit microseconds from 1601, iOS CoreData seconds from 2001, mail headers carry local offsets; drifting CMOS clocks compound the problem.	Record clock offsets at seizure; convert every artifact to UTC with the formula shown before building any matrix.
Short and unequal retention	NTFS records persist for months, but campus NAT and DHCP logs, CGNAT tables and basic-tier cloud audit logs commonly roll over in 7–30 days.	Issue preservation freezes during the identification phase, before deep analysis begins.
Pervasive encryption	FBE and iOS data protection, BitLocker and FileVault, TLS 1.3 payloads and SQLCipher messaging stores block passive inspection.	Preserve AFU state, capture volatile memory on live endpoints, obtain consent or recovery keys, correlate unencrypted metadata.
Jurisdiction and data sovereignty	SaaS tenants, phishing proxies and external VPS hosts sit in foreign jurisdictions that an institutional mandate cannot compel.	Use tenant-admin audit logs the institution already controls; escalate external addresses to law enforcement through CERT-In or mutual legal assistance channels.
Cloud sync contamination	Browser profile sync and drive clients replicate history and files onto every signed-in device, so remote attacker actions can appear on a local disk.	Inspect originator_cache_guid and sync tables; verify against local network flow records such as SRUM and NetFlow.
Identity abstraction	Shared campus NAT addresses, carrier CGNAT and VPN or Tor routing sever any direct IP-to-person mapping.	Pivot on internal DHCP leases, 802.1X MAC records, user-agent fingerprints and token identifiers rather than public IPs.
False attribution risk	A single cloud log line reading 'account X downloaded confidential files' can convict a phishing victim.	Require multi-source corroboration and test the token-theft hypothesis explicitly against the insider-misuse hypothesis.

2.8 Final Investigative Summary (Question 38)

Question 38: Prepare a final investigation conclusion distinguishing direct evidence, corroborating evidence, and unresolved questions.

Table 5.6: Deliverable Conclusion Matrix — Direct Evidence, Corroboration, Open Questions

Evidentiary Tier	Synthesized Findings (Hypothetical Case)
Tier 1 — Direct evidence	Cloud IAM and API audit logs prove account stu_aravind authenticated with token tok_9f3c at 14:14:29 UTC from 203.0.113.88 using python-requests on Linux and exported 14 grant PDFs. Mail and messaging headers prove a spoofed SSO link from 198.51.100.74 reached the student at 14:02–14:03 UTC. Mobile browser and authenticator logs prove he clicked that link at 14:05:40 UTC and approved push req-77b2 at 14:06:12 UTC, yielding the token later replayed.
Tier 2 — Corroborating evidence	Laptop event logs, $MFT, $UsnJrnl and browser stores place continuous local coursework during the window and show no grant PDF downloaded, opened or deleted; campus NetFlow and DNS logs independently show zero flows from 172.16.8.112 to the portal; the user-agent mismatch between the intrusion (Linux python-requests) and the student's devices (Windows Chrome, Android Chrome) indicates an automated external host.

Evidentiary Tier	Synthesized Findings (Hypothetical Case)
Tier 3 — Unresolved questions	The operator behind 198.51.100.74 and 203.0.113.88 is unidentified, requiring provider process. What the external host did with the 14 exported PDFs after 14:21:48 UTC is unknown. Out-of-band collusion cannot be excluded by institutional artifacts alone, since the absence of local evidence proves only the absence of local technical traces.

Summary statement: correlation across all six domains shows why single-source inference is dangerous. Had the institution relied only on the portal access log naming stu_aravind as downloader, a phishing victim would have been disciplined for theft. The evidence instead supports an external OAuth token-theft intrusion enabled by social engineering, with the student's devices acting as the credential source rather than the exfiltration path. Recommendations follow directly — phishing-resistant FIDO2 rather than push approval, conditional access on impossible travel and user-agent anomaly, tenant audit-log retention beyond 30 days, and prompt preservation requests when an incident is reported.

3. ETHICAL AND LEGAL CONSTRAINTS ON THIS SCOPE

Examining a student's personal phone and messaging records requires explicit informed consent or appropriate legal process, and scope minimization under the Digital Personal Data Protection Act, 2023. Extraction here is confined to 13 Oct 2026, 13:30–15:00 UTC; every export is hashed with SHA-256, custody is logged, and electronic records intended for adjudication are accompanied by the certification contemplated by Section 63 of the Bharatiya Sakshya Adhiniyam, 2023. Third-party data encountered incidentally in shared conversations is neither extracted nor reported.

4. REFERENCES

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